

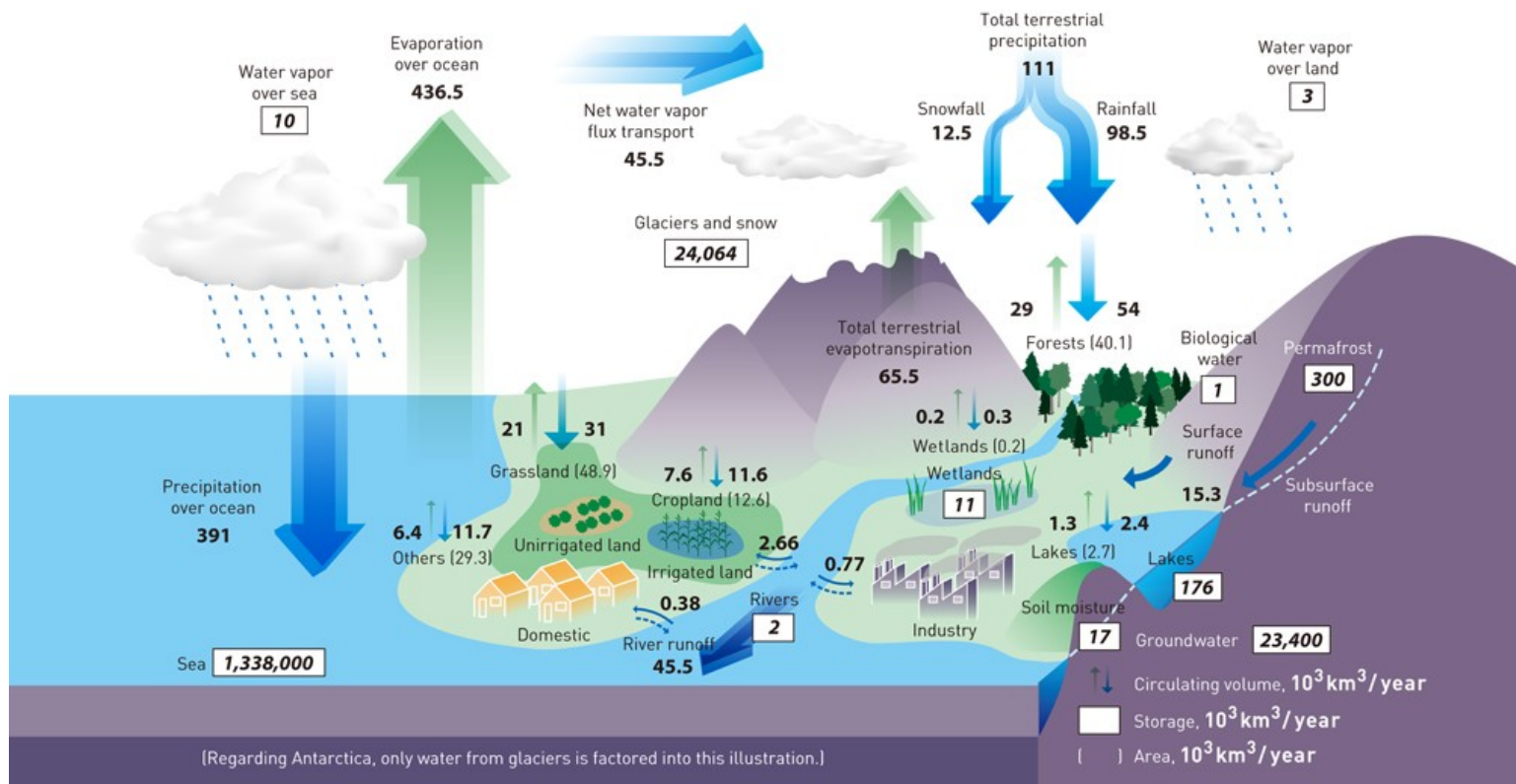
Introduction to Earth System

- Humidity
- Virtual temperature

1 page: A4, two sides (final exam)

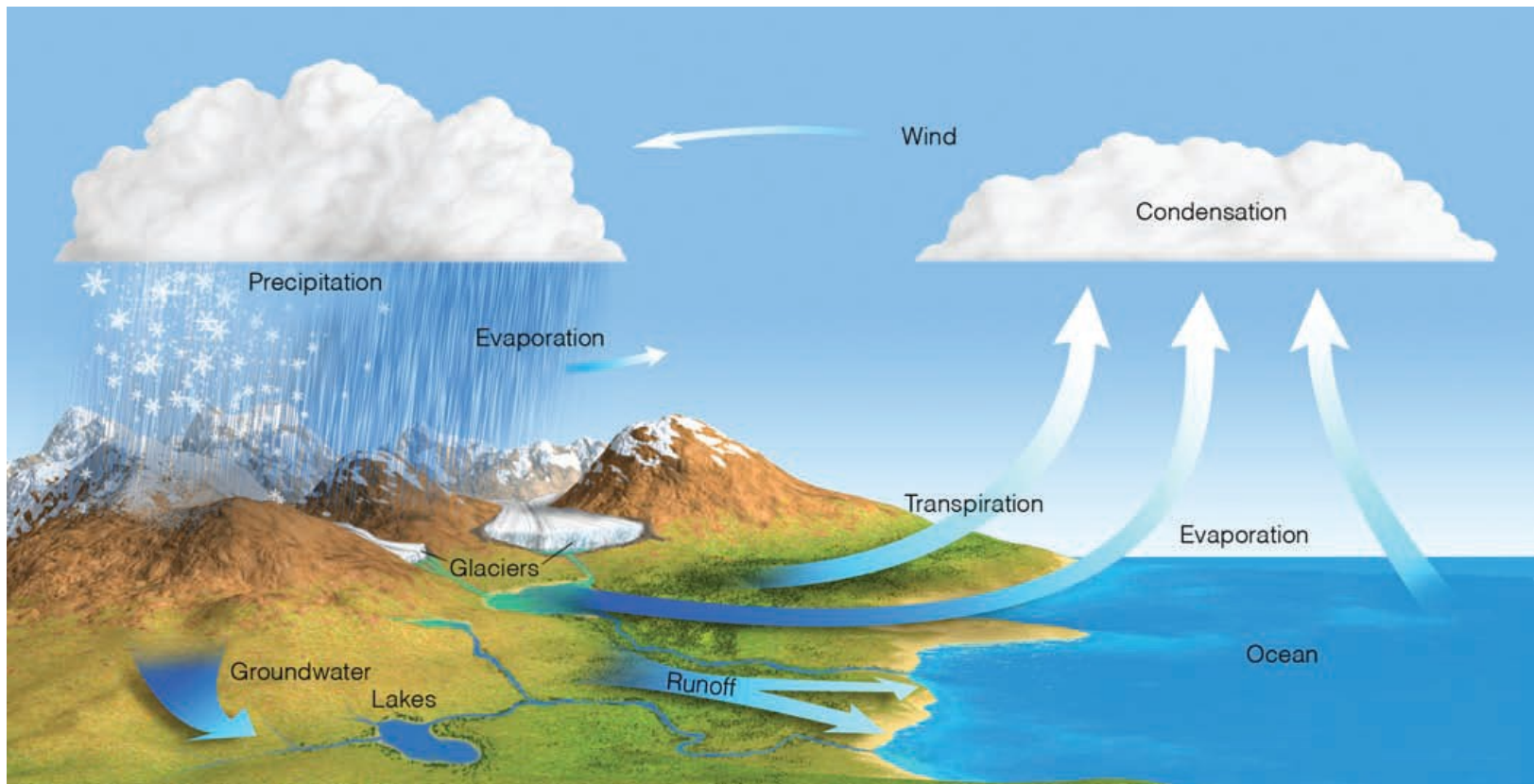
I/ Humidity

- Almost all the water in the atm. is contained within the troposphere.
- Most is in the form of water vapour, with some as cloud water or ice.
- Indispensable to life → essential role as thermal regulator & filter
- **Hydrologic cycle**

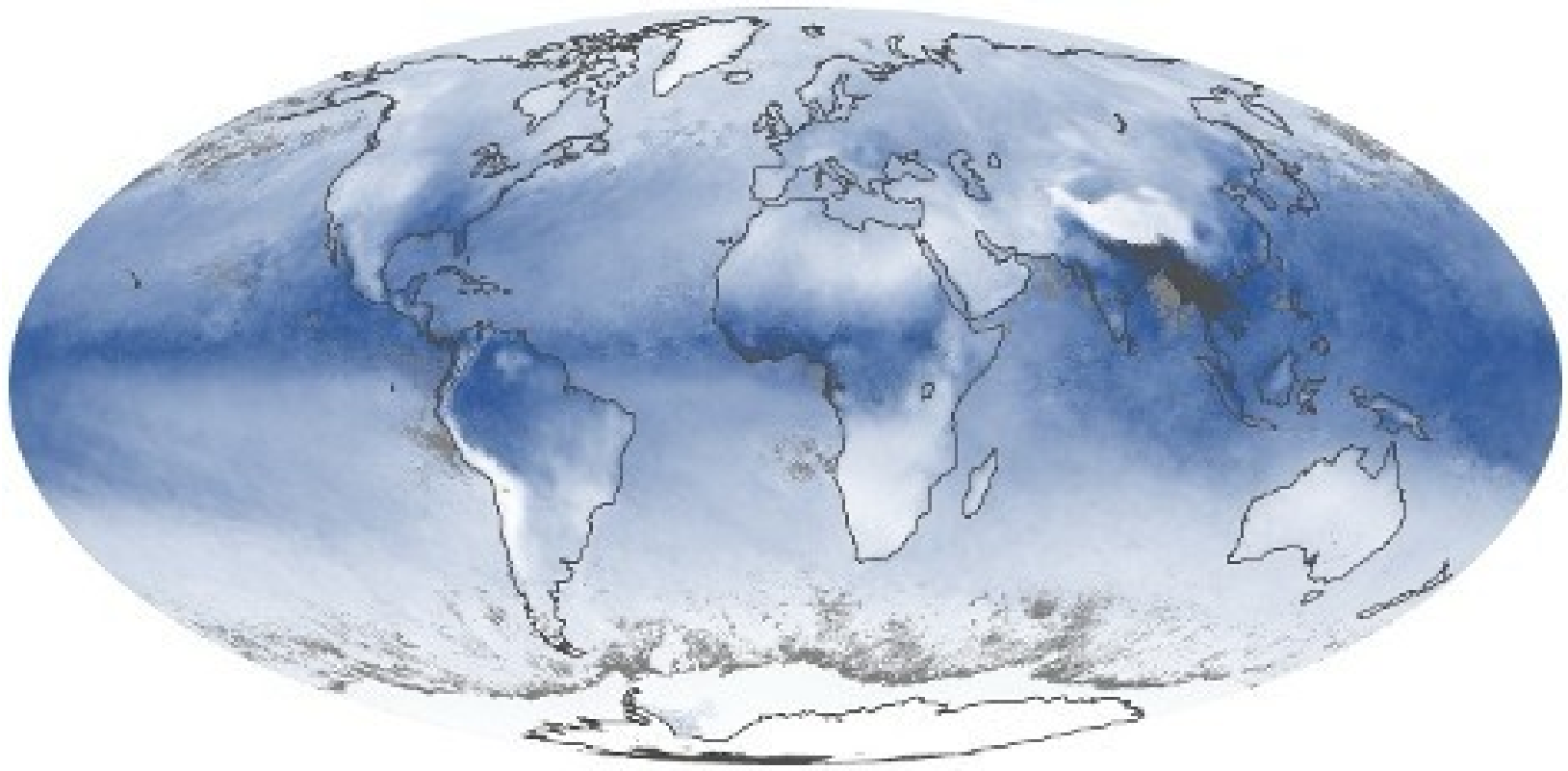


Water in the Atmosphere

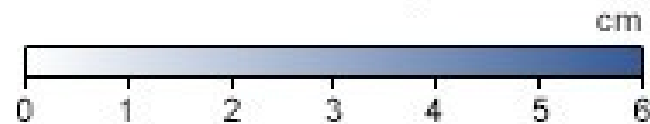
- ? cm liquid water equivalent 2.5 cm
- ? % of atmospheric mass 0.25% (including 1% in condensed form)
- ?% of global water budget 0.035% (i.e. 13 000km³)



Water in the Atmosphere



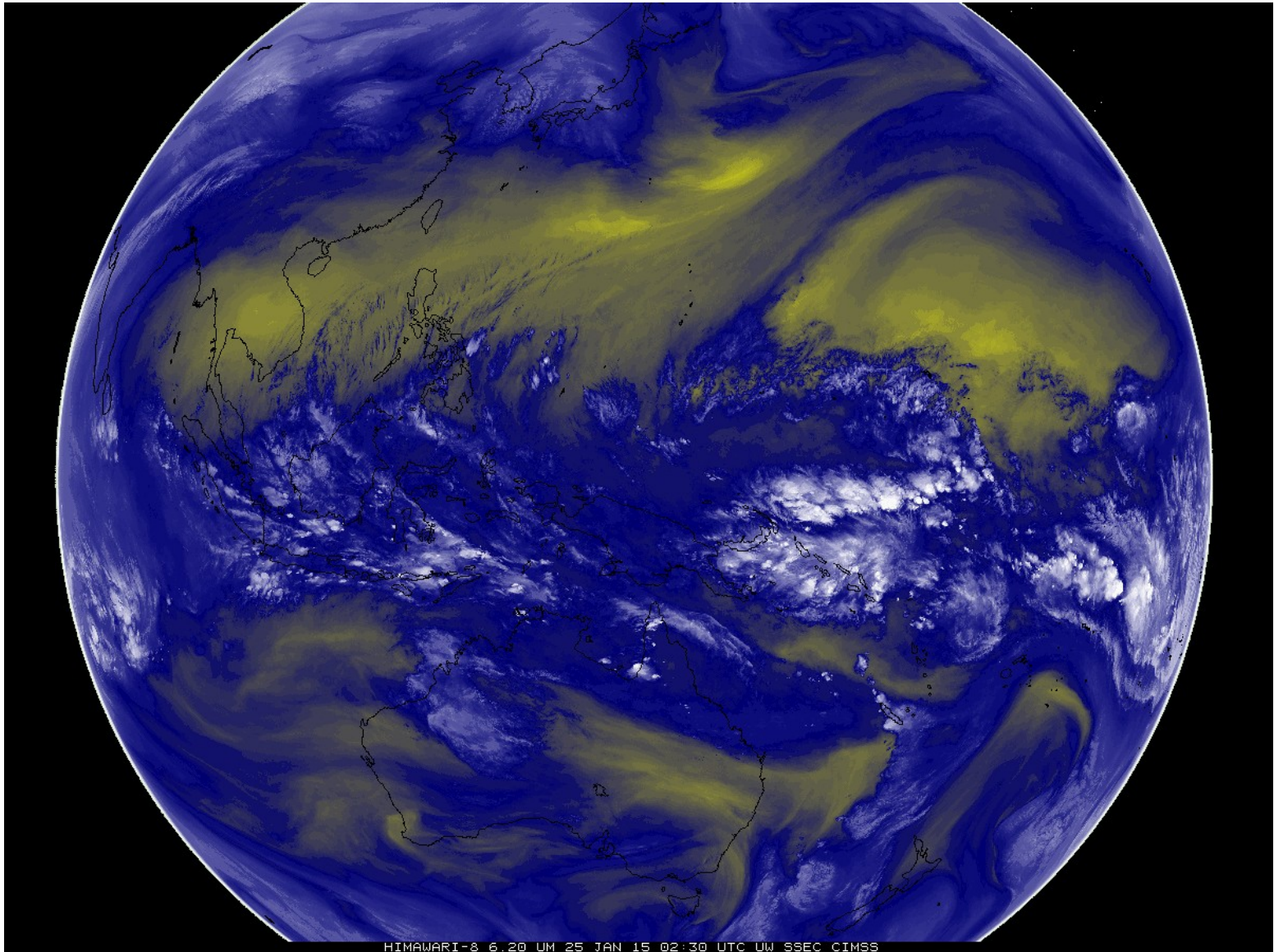
Water Vapor



July 2002

Himawari water vapor

Which band?



Buoyancy Effects

- Water in the atmosphere has important effects on dynamics, primarily convective processes
- ? Is humid air heavier than dry air?
- ? molecular weight of water: 18 g mol^{-1}
- ? mean molecular weight of dry air: $\approx 29 \text{ g mol}^{-1}$

Mole per mole comparison:

$$\rho_{\text{water vapour}} = 0.62 \rho_{\text{air}}$$

⇒ A mixture of humid air is less dense than dry (or less humid) air at the same temperature and pressure

Humidity: some basic definitions

- **Q: Absolute humidity?**

- mass of water vapor in a fixed volume of air; $AH = \frac{m_v}{V}$

- **Q: Specific humidity?**

- mass of water vapor in a fixed total mass of air; $q = \frac{m_v}{m_v + m_d}$

Exercise 1

A weather station samples a room-sized volume of air measuring 40 m^3 . After extracting all the moisture from the sample, the collected water has a mass of 640 g .

Q. Calculate the absolute humidity of the room?

Exercise 2

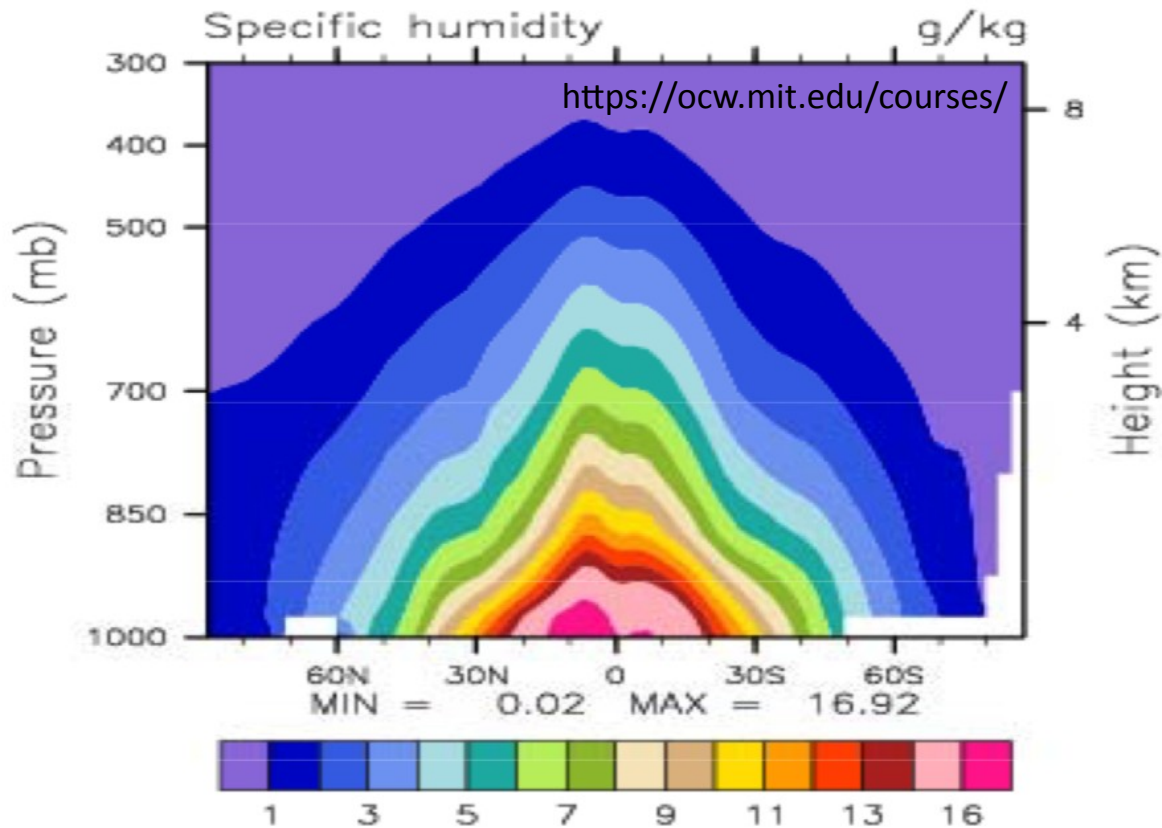
An air parcel in the lower atmosphere has a total mass of 2.5 kg. A laboratory analysis determines that this parcel contains 35 g of water vapor, while the rest is dry air.

Q. Calculate the specific humidity of this air parcel in g/kg?

Exercise 3

- Two distinct air parcels were mixed together in a sealed container:
 - **Parcel A**: has a total mass of 10 kg and a specific humidity of 8 g/kg.
 - **Parcel B**: has a total mass of 15 kg and a specific humidity of 12 g/kg.
- Q. Find the specific humidity of the resulting 25 kg mixture?

Annual mean: Specific humidity



- Higher humidity in the tropics
- Humidity decreases with altitude

Vapor pressure

- **Vapor pressure:** The pressure exerted by water vapor molecules in an air parcel
- **Dalton's law:** the total pressure exerted by a mixture of gases that do not interact chemically is equal to the sum of the partial pressures of the gases

Total atmospheric pressure

Dry air pressure

Vapour pressure

$P = P_a + e$

Dalton's law

The diagram illustrates the equation $P = P_a + e$ where P represents the total atmospheric pressure, P_a represents the dry air pressure, and e represents the vapour pressure. Arrows point from the labels to their respective terms in the equation. The label 'Dalton's law' is also present, pointing towards the equation.

Mixing ratio

- mass of water vapor in a fixed mass of dry air

$$r = \frac{m_v}{m_d}$$

- m_v : water vapor mass; m_d : dry mass for a given volume of air
- Temperate regions: $e \sim 15$ hPa ($r \sim 10$ g/kg)
- Very cold continental regions: $e \sim 1.5$ hPa ($r \sim 1$ g/kg)
- Equatorial regions: $e \sim 30$ hPa ($r \sim 20$ g/kg)

Humidity: some basic definitions

■ Q: Dew point?

- ✓ Temperature to which air would have to be cooled (same air pressure or moisture) for saturation to occur

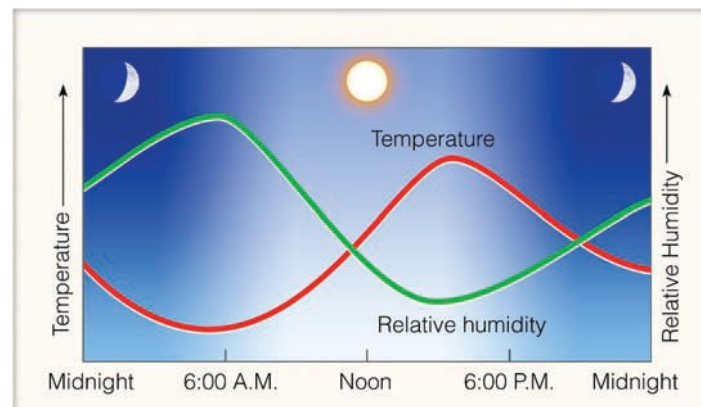
Q: High dew point indicates high or low water vapor?

■ Q: Relative humidity?

- ✓ ratio of water vapor actually in the air to the maximum amount of water required for saturation at a particular temperature and pressure
- ✓ If greater than 100 % RH →
supersaturated air



Supersaturation in the condensation trails



Exercise 4

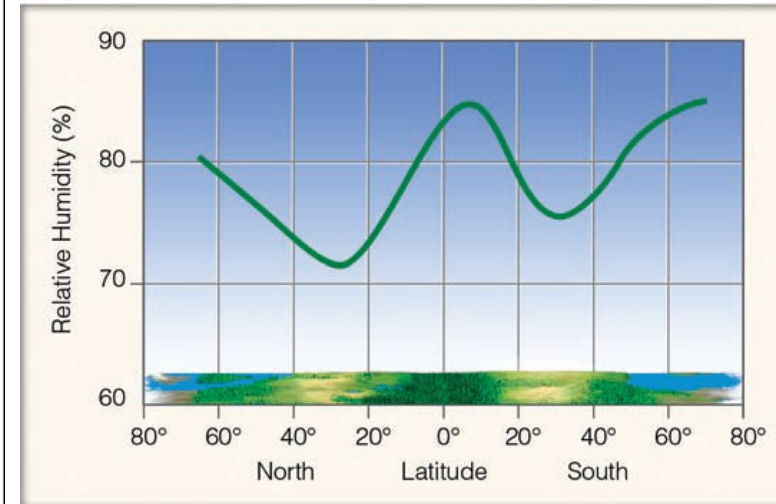
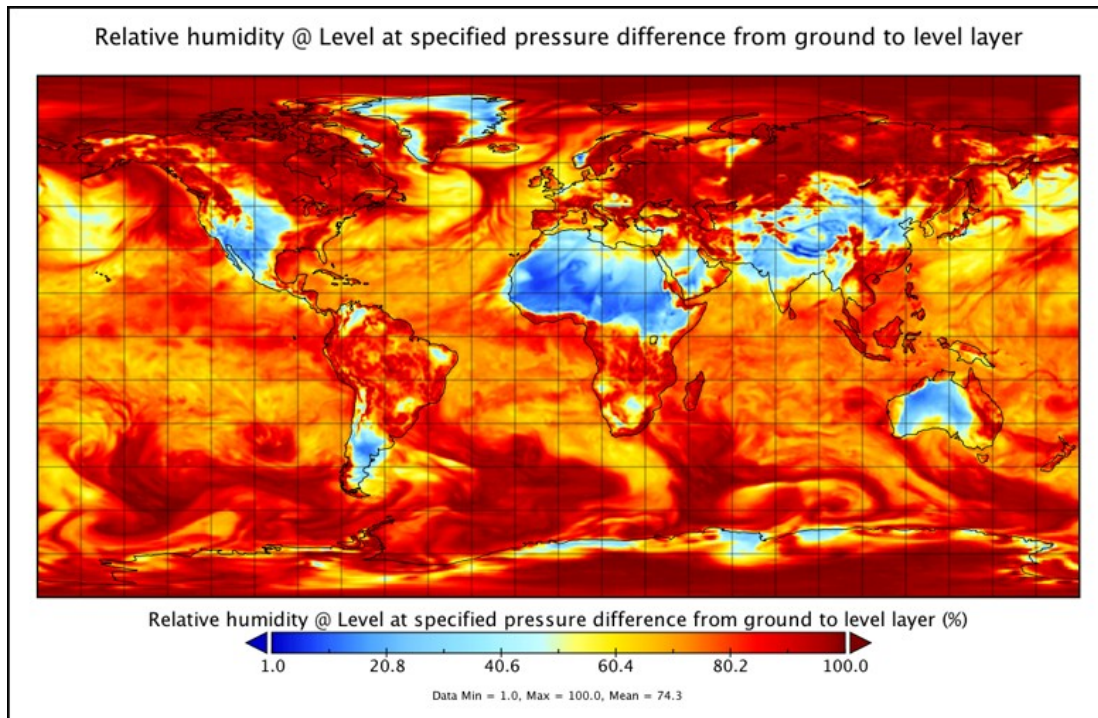
On a warm spring afternoon, the air temperature is 25°C.

A meteorologist measures the actual vapor pressure in the air to be 1.6 kPa.

Using a reference table, they find that the saturation vapor pressure e_s at 25°C is 3.2 kPa.

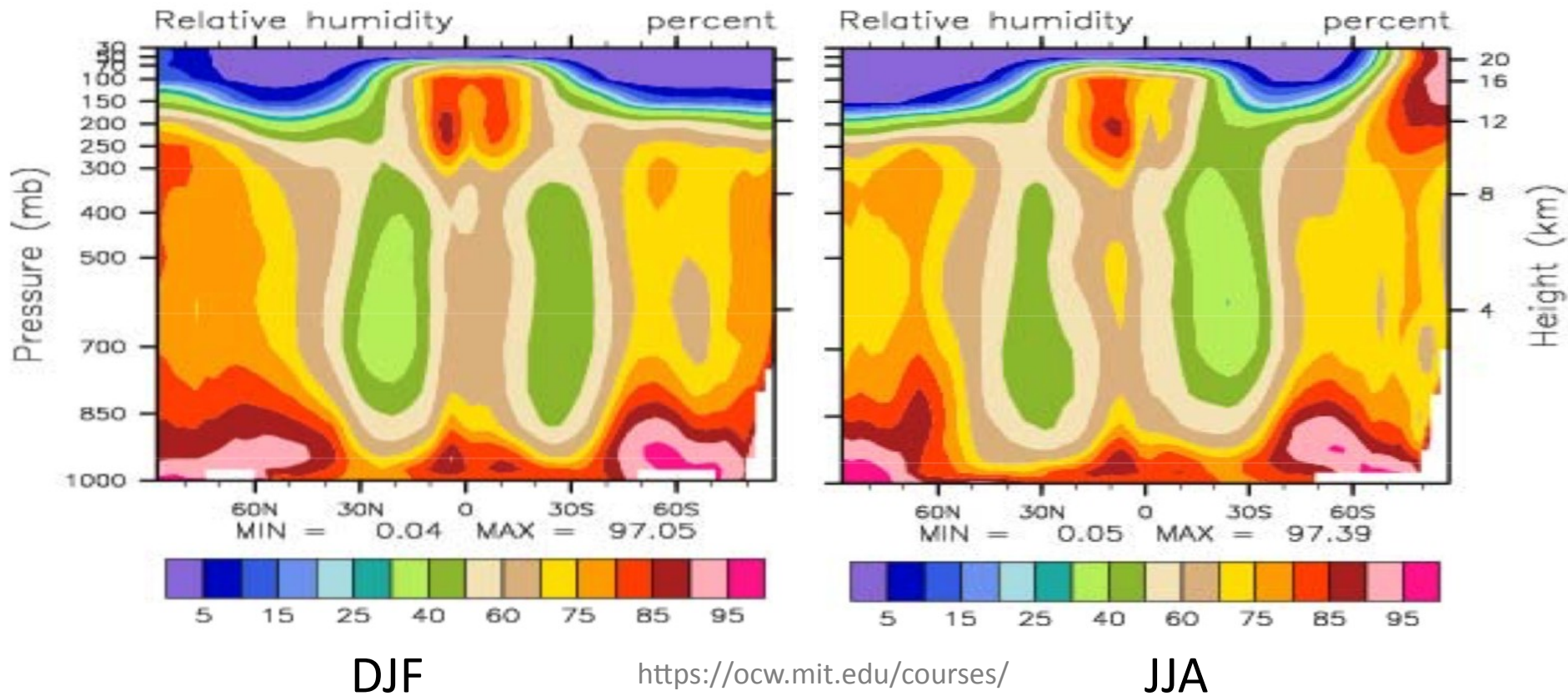
Q. Calculate the relative humidity of the air?

Relative Humidity: global distribution



- RH is high at the poles and in the tropics; i.e. temperature is close to dew point
- RH is low over deserts

Relative humidity



Q: Why relative humidity is ~100% in the high latitude areas where specific humidity is low?

Exercise 5

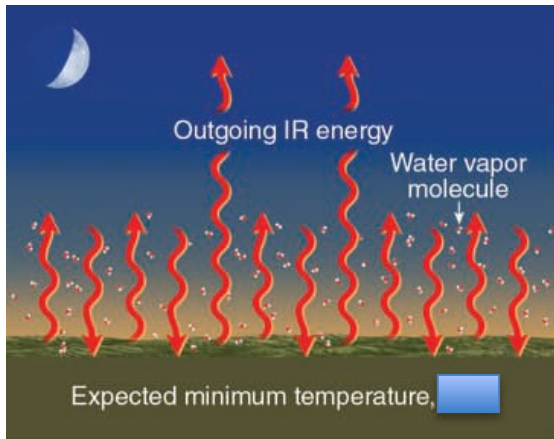
- In a place A in the polar area, temperature is -10°C , maximum water capacity (i.e. saturation vapor density) is 2 g/m^3 .
- In Cairo (desert), temperature is 35°C , maximum water capacity is 40 g/m^3 .

An air sample collected in A is found to contain 1.8 g/m^3 of water vapor.

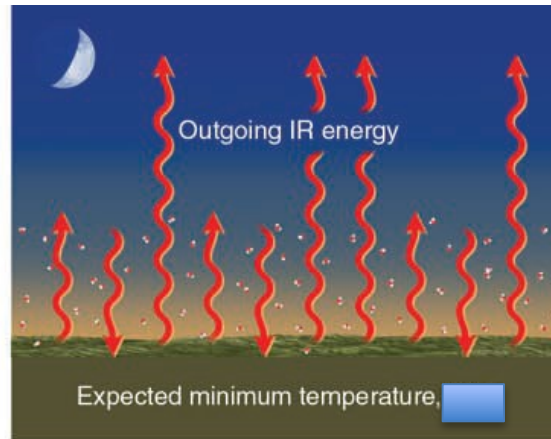
An air sample collected in Cairo is found to contain 4.0 g/m^3 of water vapor.

Q. Calculate the relative humidity of the air in A and in Cairo?

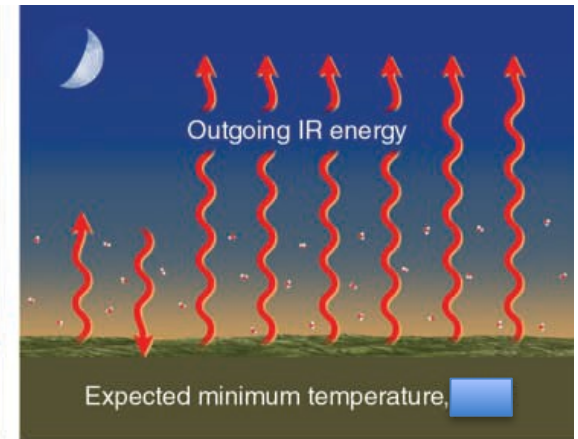
Dew point



Dew-point temperature, 21°C



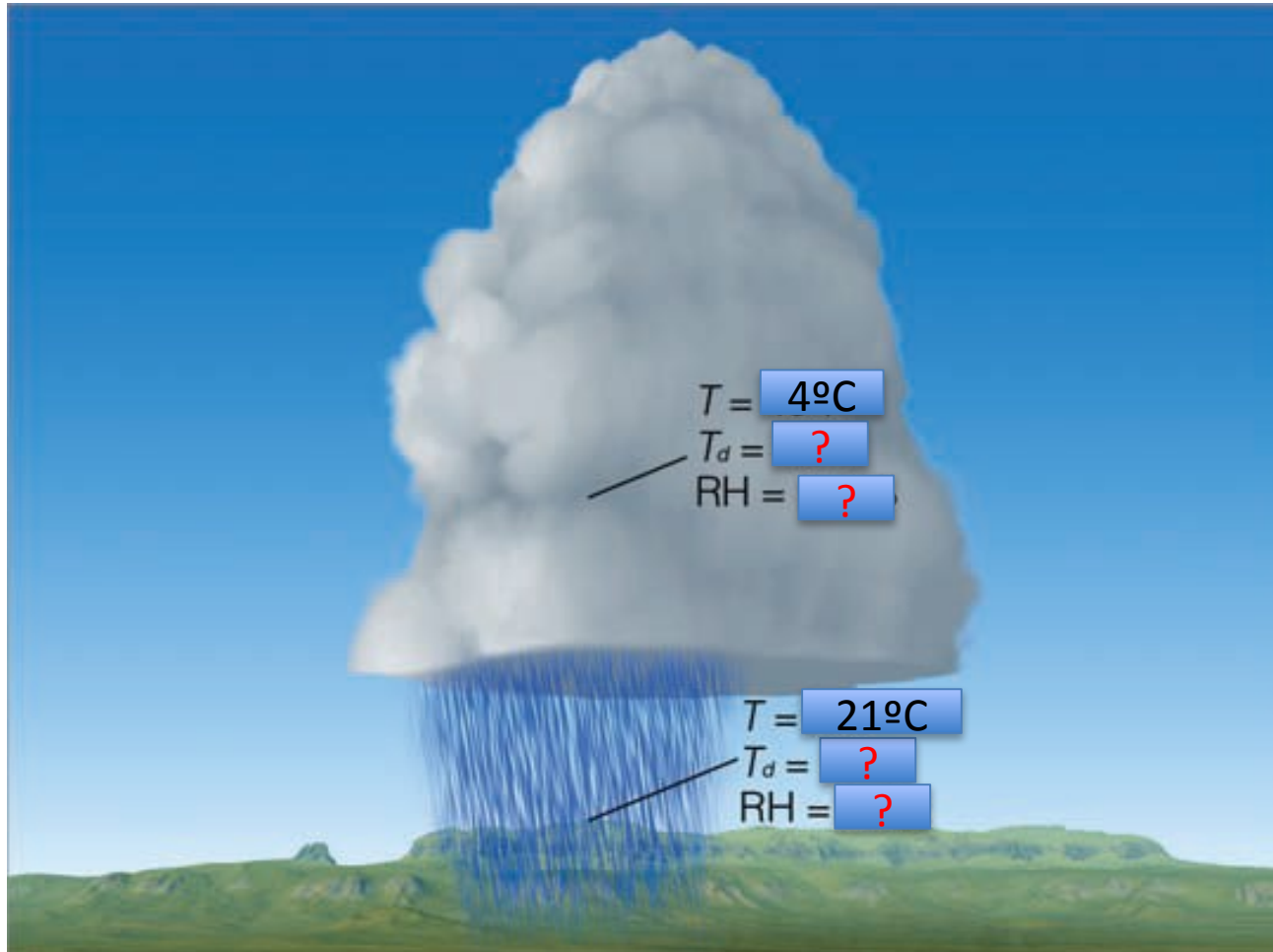
Dew-point temperature 15.5°C



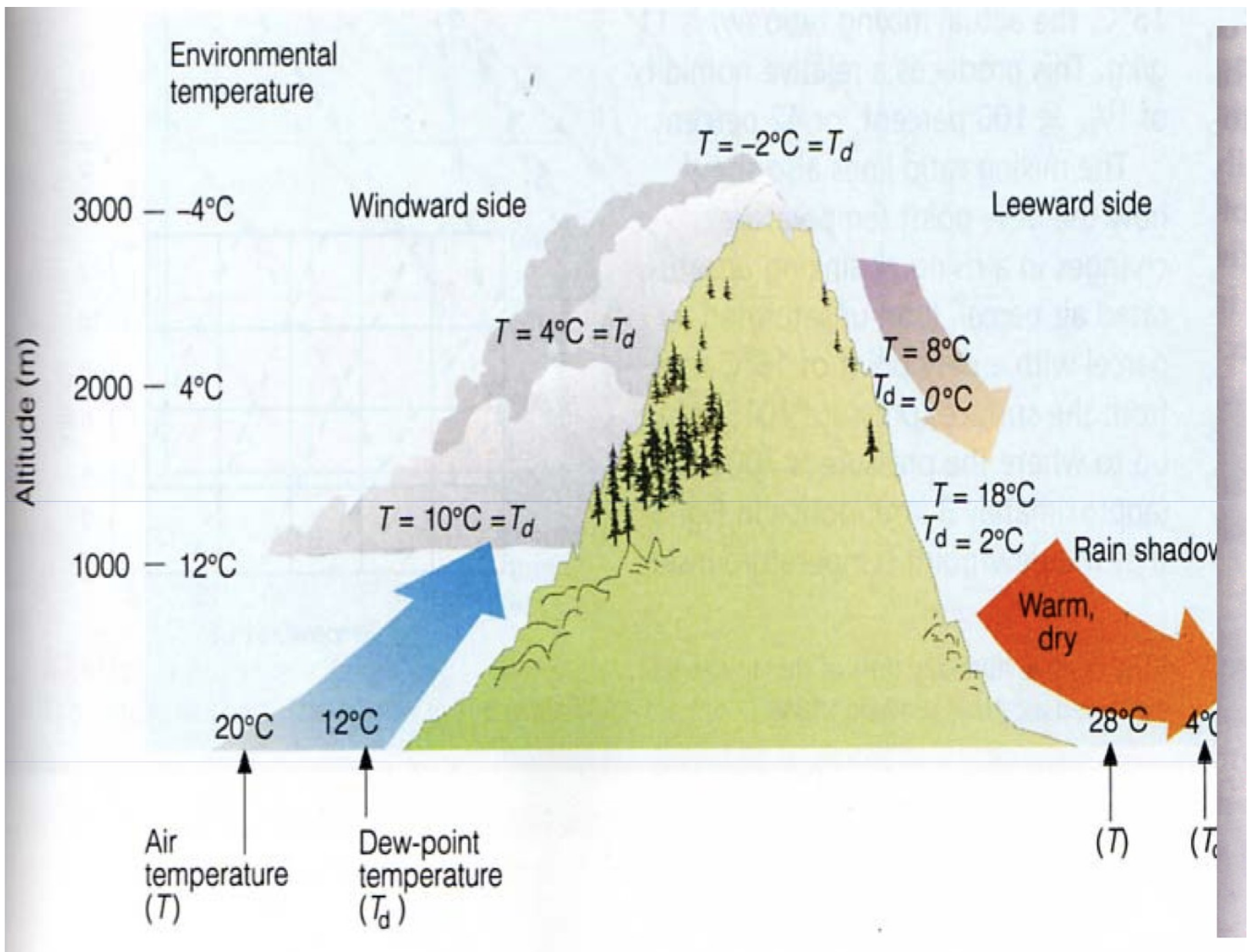
Dew-point temperature 10°C

- On a calm, clear night, the lower the dew-point temperature →? the **lower | higher** the expected minimum temperature. **Why?**
- **Explanation**
lower dew point → less water vapor in the air to absorb and radiate infrared energy back to the surface → More IR energy from the surface is able to escape into space → **more rapid radiative cooling** at the surface.

RH & dew point



Inside the cloud?
At the surface?



II/ Virtual temperature

- **The perfect gas equation:** $p = \rho R T$ (1)

- Dry air $\rightarrow R_d = 287 \text{ J kg}^{-1} \text{ K}^{-1} \rightarrow p_d = \rho_d R_d T$

- Same pressure $\rightarrow$ having moisture reduces the overall density

$\rightarrow$ **Virtual temperature:** temperature that dry air would need to reflect the same pressure as the given moist air sample at the same density

$$p = \rho R_d T_v \quad (2) \quad \text{Q. Compare } T_v \text{ and } T?$$

Specific humidity $q = \frac{\rho_v}{\rho_d + \rho_v}$

Mixing ratio $w = \frac{\rho_v}{\rho_d}$

We can derive the virtual temperature :

$$\rightarrow T_v = T [1 + 0.61 q] \quad (3) \quad \text{Q. How to obtain this equation?}$$

Quizz

- **Q:** two air parcels have the same size, the same pressure and temperature, but one is dry and the other is moist air. Which one is less dense?

A: moist air as dry air is 29g/mol, water vapor is 18 g/mol

- **Q:** Moist air has a higher or lower virtual temperature than dry air that has the same temperature?

A: higher T_v (because lower density)

- **Q:** Consider an air parcel ($T=25^\circ\text{C}$, $q=10\text{g/kg}$) at the same pressure level as a surrounding environment ($T_{\text{env}}=26^\circ\text{C}$, $q_{\text{env}}=1\text{g/kg}$). Does the air parcel rise?

$$T_{v_{\text{parcel}}} = (25 + 273) * (1 + 0.61 * 0.01 \text{ kg/kg}) = 299.8\text{K}$$

$$T_{v_{\text{env}}} = (26 + 273) * (1 + 0.61 * 0.001 \text{ kg/kg}) = 299.2\text{K}$$

$p = \rho R_d T_v \rightarrow$ the air parcel is less dense than its environment $\rightarrow$ rising.

Practice #9

- Download temperature and specific humidity from pressure levels of the ERA5 data:

https://remosat.usth.edu.vn/~thanhnd/Download/tmp/ERA5_S H_Temp.nc

- Open the file and read the data
- Compute virtual temperature at each level
- Plot the 2D map (latitude & altitude) of virtual temperature and the difference between T_v and T at each level
- Comment on the location of potential instability.